



David Tolpin

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Hands-on Scientist / Applied ML Researcher

Applied machine learning researcher and computer scientist with experience spanning industry R&D, production systems, probabilistic modeling, human motion, computer vision, time-series modeling, anomaly detection, planning, and probabilistic programming. Strong preference for roles that combine research-level problem formulation with direct implementation, experimentation, deployment, and collaboration with engineering teams.

Recent work focuses on computer vision and graphics algorithms, 3D human motion, SMPL pose modeling, neural inverse kinematics, pose priors, differentiable 3D modeling, and coding automation harnesses. Earlier work includes forecasting and decision-making systems for online advertising, deterioration prediction in clinical time series, cybersecurity anomaly detection, probabilistic programming, Bayesian inference, reinforcement learning, and heuristic search.

Technical Focus

Machine learning and statistics: Bayesian modeling, probabilistic inference, density estimation, normalizing flows, time-series modeling, anomaly detection, forecasting, reinforcement learning, planning under uncertainty. **Motion / vision / graphics:** markerless motion capture, SMPL human models, pose priors, inverse kinematics, differentiable 3D modeling, computer vision and graphics algorithms. **Systems and implementation:** production research code, distributed data processing, Go, Python, Julia, Clojure, Scala, Unix tooling, reproducible CLI workflows. **Research practice:** problem formulation, experiment design, papers, patents, algorithm design, scientific collaboration, mentoring and technical leadership.

Experience

2K — Computer Scientist

September 2025 – Present

Designing computer vision and graphics algorithms and automating coding harnesses.

Selected work:

- Implemented transfer of garment movements from 2D AI-generated video clips to 4D (rigged mesh model) motion sequences. Applied 5D (RGB+XY) optimal-transport differentiable rendering for the task on top of PyTorch and nvdiffrast.
- Created a self-persisting agent-based framework for prompt-driven authoring and editing of multi-character scenes with physical and intentional interactions. Built the framework around Pi as the agent REPL. Applied compounding engineering principles by turning interactively discovered workflows into reusable agent instructions, skills, tool prompts, and verification procedures.

YOOM (acquired by 2K) — Algorithm Engineer

May 2023 – September 2025

Worked on 3D human modeling and markerless motion-capture problems.

Selected work:

- Trained a keypoints-to-rotations transformer for a rigged human body model, serving as a time-efficient alternative to coarse motion capture optimization. Helped to adapt the model to studio-deployed online motion capture.
- Trained a RealNVP-based neural human pose prior with SOTA performance and helped deploy the prior for markerless motion capture.
- Applied principles of mixture models to help implement efficient and robust markerless motion capture of multi-character scenes.
- Designed and implemented algorithms for robust and efficient transfer of 2D character faces onto a rigged mesh head model. Coupled the implementation with an AI image generator which facilitated cost-efficient generation of diverse high-quality personalities for games.

Wild Biotech — Machine Learning Scientist

November 2022 – April 2023

Designed and implemented ML and inference methods for computational biology.

Selected work:

- Designed and coded, with colleagues, a machine learning and inference architecture for discovery and scoring of miniprotein–protein interactions.
- Invented, evaluated, and deployed a state-of-the-art method for post-processing and analysis of phage-display NGS results.
- Helped the team adopt modern software development techniques, tools, and libraries.
- Improved information flow and technical coordination in a distributed hybrid team.

Ben-Gurion University of the Negev — Senior Lecturer / Assistant Professor

October 2019 – October 2025

Research and teaching in probabilistic programming, Bayesian statistical modeling, AI planning, and reinforcement learning.

Selected work:

- Conducted research in probabilistic programming, Bayesian inference, planning, and reinforcement learning. Published work on planning as policy inference, Bayesian inference reuse in multilevel models, probabilistic programs as action descriptions, stochastic conditioning, and related topics.
- Supervised and collaborated on research bridging probabilistic modeling, AI planning, and implementable inference systems.
- Authored and taught courses “Mathematical foundations of computer science” and “Applied Bayesian data analysis”.

PubPlus — Chief Data Scientist

May 2018 – September 2022

Led data science and statistical modeling for advertising campaign optimization, revenue attribution, and forecasting.

Selected work:

- Designed and helped implement a distributed architecture for data collection and processing.
- Designed forecasting algorithms that became the basis for automated decision-making in advertising campaign management and user acquisition.
- Used rigorous statistical modeling to discover and mitigate problems in complex data sources.
- Introduced Go into the company’s software stack and integrated ML/statistical modeling tasks into server-side Go services, partly using libraries written in-house.

CLEW Medical — Computer Scientist

August 2017 – March 2018

Worked on Bayesian inference and deep learning for clinical time-space models and early deterioration prediction in intensive care.

Selected work:

- Introduced a scoring scheme for deterioration-prediction evaluation and training.
- Designed and implemented a changepoint-detection algorithm for multi-variate vital-sign time series.
- Led adoption of deep-learning tools and techniques for next-generation deterioration-prediction algorithms.

PayPal — Research Scientist

August 2015 – July 2017

Applied research in anomaly detection, cybersecurity, and data science.

Selected work:

- Designed and implemented anomaly-detection algorithms across several domains.
- Wrote approximately 20,000 lines of production/research code in Python, Scala, Lua, C, and Clojure.
- Built a data science team of four scientists.
- Led scientific collaboration with Ben-Gurion University.
- Submitted nine patent applications, contributed to two open-source projects, and co-authored four academic papers.

University of Oxford — Post-Doctoral Researcher

August 2014 – July 2015

Worked on probabilistic programming, including the Anglican probabilistic programming system, inference algorithms, and applications of probabilistic programming to AI.

Selected work:

- Developed probabilistic-programming infrastructure and inference algorithms.
- Contributed to Anglican, including work later represented in highly cited publications on probabilistic programming.
- Applied probabilistic programming to AI and decision-making problems.

Earlier Experience

Ben-Gurion University of the Negev — Postdoctoral Researcher / Lecturer / PhD Student Research in rational metareasoning and heuristic search; taught programming languages, systems, computer architecture, compiler construction, and related courses.

RenderX — CTO Designed and led implementation of one of the first commercial XSL Formatting Objects engines; wrote core modules of the formatting engine.

Polimetrix (acquired by YouGov) — Consultant Designed and developed a high-volume online survey system with flexible tools for survey designers.

IREX — IATP Armenia Coordinator Established a network of public internet access points in Armenian universities and libraries; implemented web mail and hosting services; taught internet literacy and web design.

Selected Publications

- **MDP Planning as Policy Inference**, arXiv 2026.
 - **Bayesian Inference Reuse in Multilevel Models**, TOPML 2026.
 - **Neural Human Pose Prior**, arXiv 2025.
 - **Fast Neural Inverse Kinematics on Human Body Motions**, * arXiv 2025.
 - **Probabilistic programs with stochastic conditioning**, ICML 2021.
 - **Deployable Probabilistic Programming**, Onward! 2019.
 - **Design and implementation of probabilistic programming language Anglican**, IFL 2016.
 - **Selecting computations: Theory and applications**, UAI 2014.
 - **MCTS based on simple regret**, AAAI 2012.
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Patents

Multiple patents and patent applications across machine learning, cybersecurity, malware detection, fraud prediction, rendering, and document formatting.

Education

PhD, Computer Science — Ben-Gurion University of the Negev, 2010–2013
MSc, Computer Science — Ben-Gurion University, 2007–2009

Languages

English, Hebrew, Russian, Yiddish, Armenian.